

Foundation (Jalapeno)

- Q1.** What is the decimal equivalent of the fraction $\frac{3}{4}$?
(A) 0.25
(B) 0.5
(C) 0.75
(D) 0.8
(E) 0.9
- Q2.** A spacecraft is traveling at $\frac{3}{5}$ of the speed of light. What percent of the speed of light is this?
(A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q3.** What is the fraction equivalent of the decimal 0.6?
(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{2}{3}$
(D) $\frac{3}{5}$
(E) $\frac{3}{4}$
- Q4.** A lab experiment requires a solution that is 25% acidic. If the solution is currently $\frac{1}{5}$ acidic, what operation must be performed to achieve the desired acidity?
(A) Add 20% acid
(B) Add 15% acid
(C) Remove 5% acid
(D) Remove 10% acid
(E) Remove 20% acid
- Q5.** The planet Mars has a radius of approximately $\frac{3}{4}$ that of Earth. If Earth's radius is 6,371 kilometers, what is Mars' radius in kilometers?
(A) 3,000
(B) 4,000
(C) 4,700
(D) 5,000
(E) 5,500
- Q6.** A space telescope is able to resolve objects at a distance of 0.8 light-years away with $\frac{3}{4}$ accuracy. How accurate is the telescope at a distance of 1 light-year?
(A) $\frac{1}{2}$
(B) $\frac{2}{3}$
(C) $\frac{3}{5}$
(D) $\frac{1}{3}$
(E) $\frac{2}{5}$
- Q7.** A physics experiment involves a pendulum with a length of $\frac{2}{3}$ meters. If the pendulum is shortened by 0.1 meters, what fraction of the original length remains?
(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{2}{5}$
(D) $\frac{4}{5}$
(E) $\frac{5}{6}$
- Q8.** A star is 3.5 light-years away from Earth. If a spacecraft travels at $\frac{3}{5}$ of the speed of light, how long will it take to reach the star?
(A) 5 years
(B) 6 years
(C) 7 years
(D) 8 years
(E) 9 years

Open Ended Questions (FRQ)

- Q9.** A spacecraft is traveling through the galaxy, and its fuel tank is $\frac{3}{4}$ full. If 0.25 of the fuel is used to travel to the nearest planet, what fraction of the fuel remains? Show your work and explain your answer.
- Q10.** A lab experiment requires a mixture of two substances, A and B. If substance A makes up 40% of the mixture and substance B makes up 30%, what fraction of the mixture is made up of other substances? Show your work and explain your answer.

Chef's Tips

- Ensure students show their work for all calculations
- Allow students to use calculators for complex calculations
- Encourage students to check their answers for reasonableness